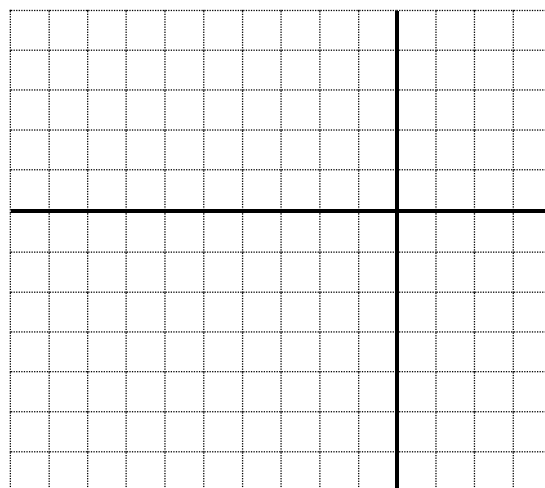
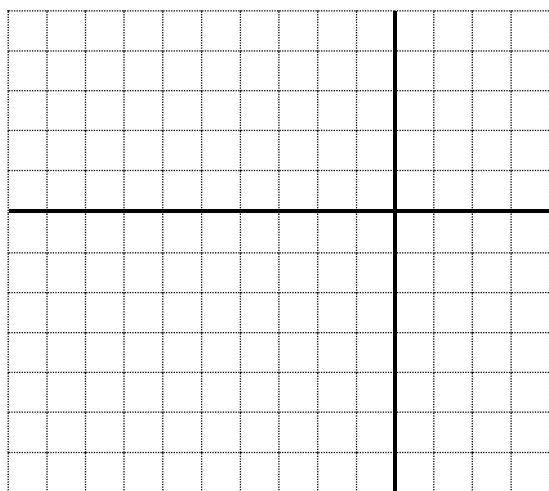


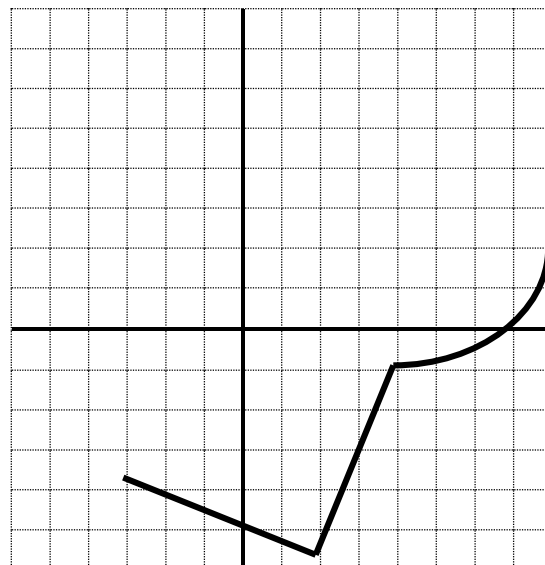
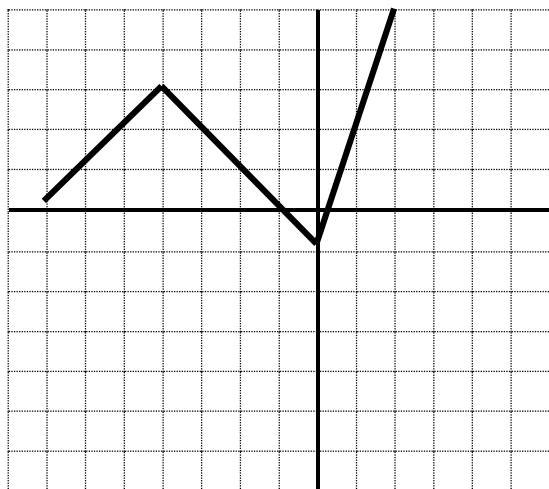
1. Graph of the following functions, labeling at least three major points on each sketch.

a) $y = -2\left(\frac{1}{2}x + \frac{3}{2}\right)^2 + 2$

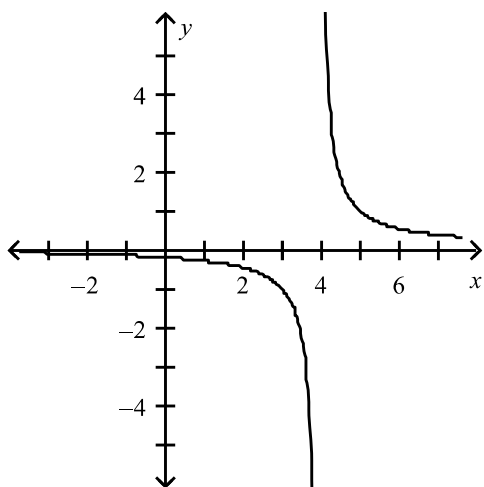
b) $y = \frac{1}{\frac{1}{2}x + 2} - 2$



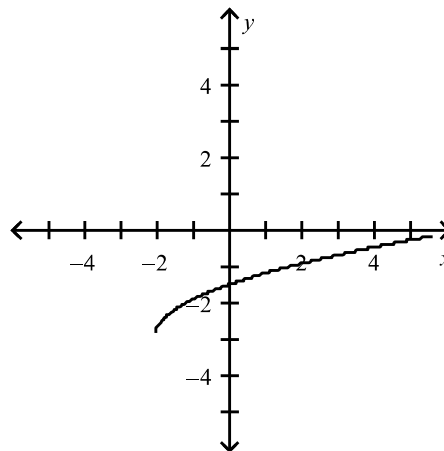
2. Graph the inverse function.



3. Find the equation of the function $f(x)$. 4. Find the equation of the function $f(x)$.



Equation: _____



Equation: _____

5. Describe the transformations that must be applied to the following:

$$f(x) = \frac{1}{x} \text{ to obtain the graph of } y = -\frac{2}{10x+5} - 7$$

6. Find the equation of the transformed functions as described below.

The graph of $y = f(x)$ has been reflected in the x-axis, horizontally compressed by a factor of one-third, shifted up two units. What is the equation of this transformed function?

Equation: _____

The graph of $y = \sqrt{x}$ has been reflected in the y-axis, horizontally stretched by a factor of two, shifted left three units. What is the equation of this transformed function?

Equation: _____

7. A function $f(x) = \{(-1, -5), (0, -1), (1, 2), (3, 10)\}$. If the function is transformed into $y = 3f(-x + 2) - 1$, state the transformed points.
8. A function $g(x) = \{(-4, -7), (-2, -3), (1, 2), (4, 6)\}$. If the function is transformed into $y = -\frac{1}{2}g(2x - 3) + 1$, state the transformed points.
9. A function $f(x) = \{(-1, -5), (0, -1), (1, 2), (3, 10)\}$ is transformed into $g(x) = af(k(x - d)) + c$ where $g(x) = \{(1, 6), (3, -2), (5, -8), (9, -24)\}$. Find the equation of $g(x)$.
10. A function $g(x) = \{(-4, -7), (-2, -3), (1, 2), (4, 6)\}$ is transformed into $g(x) = af(k(x - d)) + c$ where $g(x) = \{(9, -16), (5, -4), (-1, 11), (-7, 23)\}$. Find the equation of $g(x)$.
11. For a function of the form $y = af(k(x - d)) + c$, describe how the domain and range changes from the base graphs of
- $f(x) = x^2$
 - $f(x) = \frac{1}{x}$
 - $f(x) = \sqrt{x}$
12. Explain the relationship between a horizontal stretch $y = f(kx)$ and a vertical stretch $y = af(x)$. How are a and k related?